

Theorems:

Week 1: Theorem 1.1 Let A and B be statements. Then $A \Rightarrow B$ is equivalent to $\overline{A} \vee B$. That is, $(A \Rightarrow B) \Leftrightarrow (\overline{A} \vee B)$.

Theorem 1.2 Let A and B be statements. Then $A \Rightarrow B$ is equivalent to $\overline{B} \Rightarrow \overline{A}$. That is, $(A \Rightarrow B) \Leftrightarrow (\overline{A} \vee B)$.

Theorem 1.3 Let A and B be statements. Then $A \vee B$ is equivalent to $\overline{A} \wedge \overline{B}$.

Theorem 2.4 Given any two real numbers $a < b$, there is some $r \in \mathbb{Q}$ such that $a < r < b$.

Theorem 2.5 The number $\sqrt{2}$ is irrational.

Week 2:

Theorem 3.7 -AM-GM Inequality Let $n \in \mathbb{N}$ and $x_1, \dots, x_n \geq 0$. Then $(x_1 \cdot x_2 \cdot \dots \cdot x_n)^{1/n} \leq \frac{(x_1 + \dots + x_n)}{n}$

Theorem 3.8 -Cauchy-Schwarz Inequality Let $n \in \mathbb{N}$, $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{R}$. Then $x_1 y_1 + \dots + x_n y_n \leq$

$$\sqrt{x_1^2 + \dots + x_n^2} \sqrt{y_1^2 + \dots + y_n^2}$$

Theorem 3.8 Triangle Inequality For $x, y \in \mathbb{R}$, $|x + y| \leq |x| + |y|$.

Corollary 3.12 Reverse Triangle Inequality $||x| - |y|| \leq |x + y|$. in particular: $|x| - |y| \leq |x + y|$

Theorem 4.1 -Geometric Series Let $x \neq 1$ be a real number. Then for $N \in \mathbb{N}$, $\sum_{n=1}^N x^n = \frac{x - x^{N+1}}{1 - x}$ and $\sum_{n=0}^N x^n = \frac{1 - x^{N+1}}{1 - x}$

Theorem 4.2 Every Integer $n \geq 2$ can be written as a product of primes $n = p_1 \dots p_k$

Week 3: Completeness Axiom for the real numbers: Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Theorem 5.1 the Archimedean property holds. That is, for any $x, y \in \mathbb{R}$ with $x, y > 0$, there is some $n \in \mathbb{N}$ such that $ny > x$.